ETH zürich

Project: Privacy-Preserving Federated Learning

From Data to Solutions
ETH Zürich
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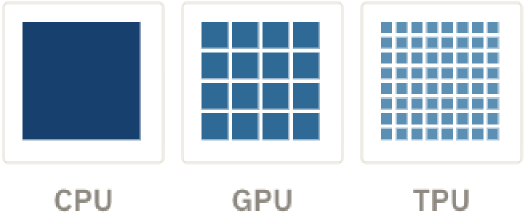
Recap: What you have learned.



same job, more units doing it

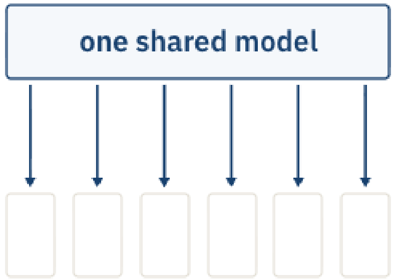
CPU, GPU, TPU

Where the work runs, and why the fastest chip sits idle.

you price the run before you start it

Training budgets

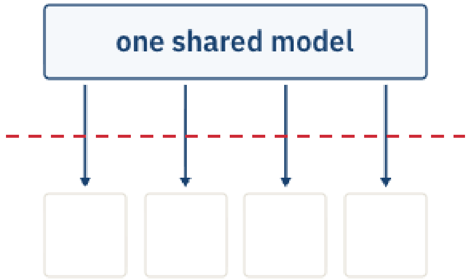
What a run costs, worked out before it starts.



machines you own

Distributed training

One model, many machines. All of them yours.

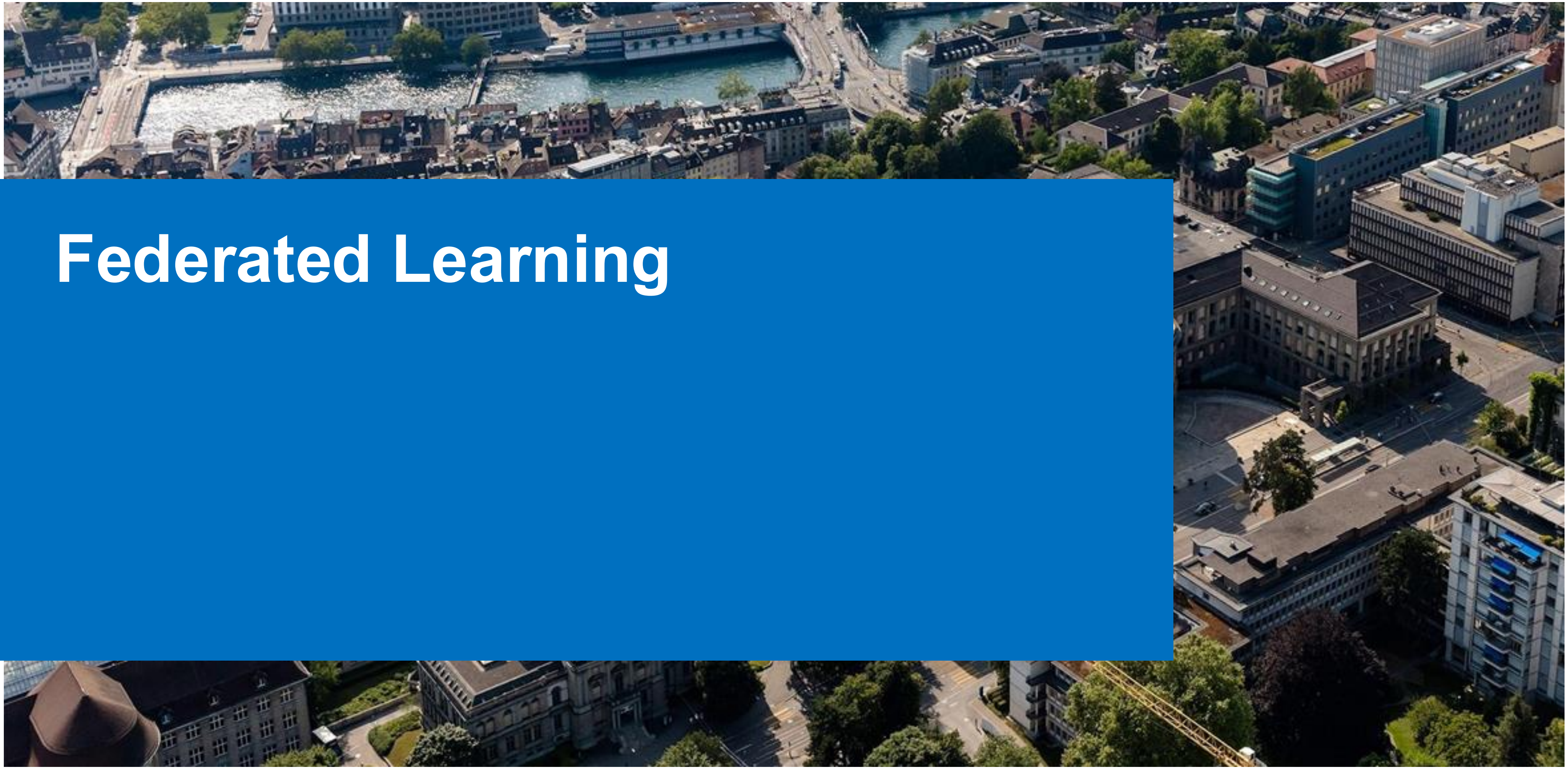


clients you do not

This project

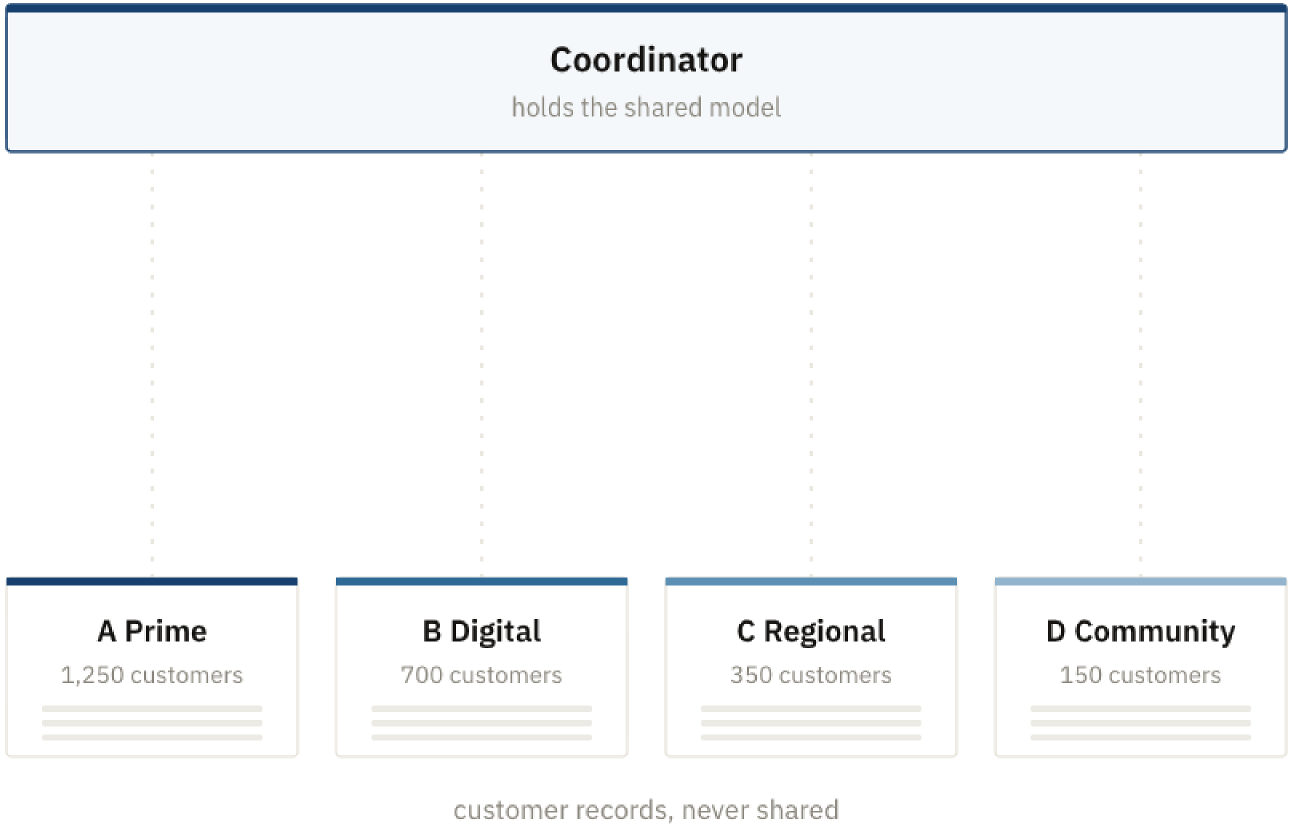
Each client has private data you cannot see.

The wall is no longer hardware. It is that the data cannot move at all.



Federated Learning

Step 1: Set up the federation.

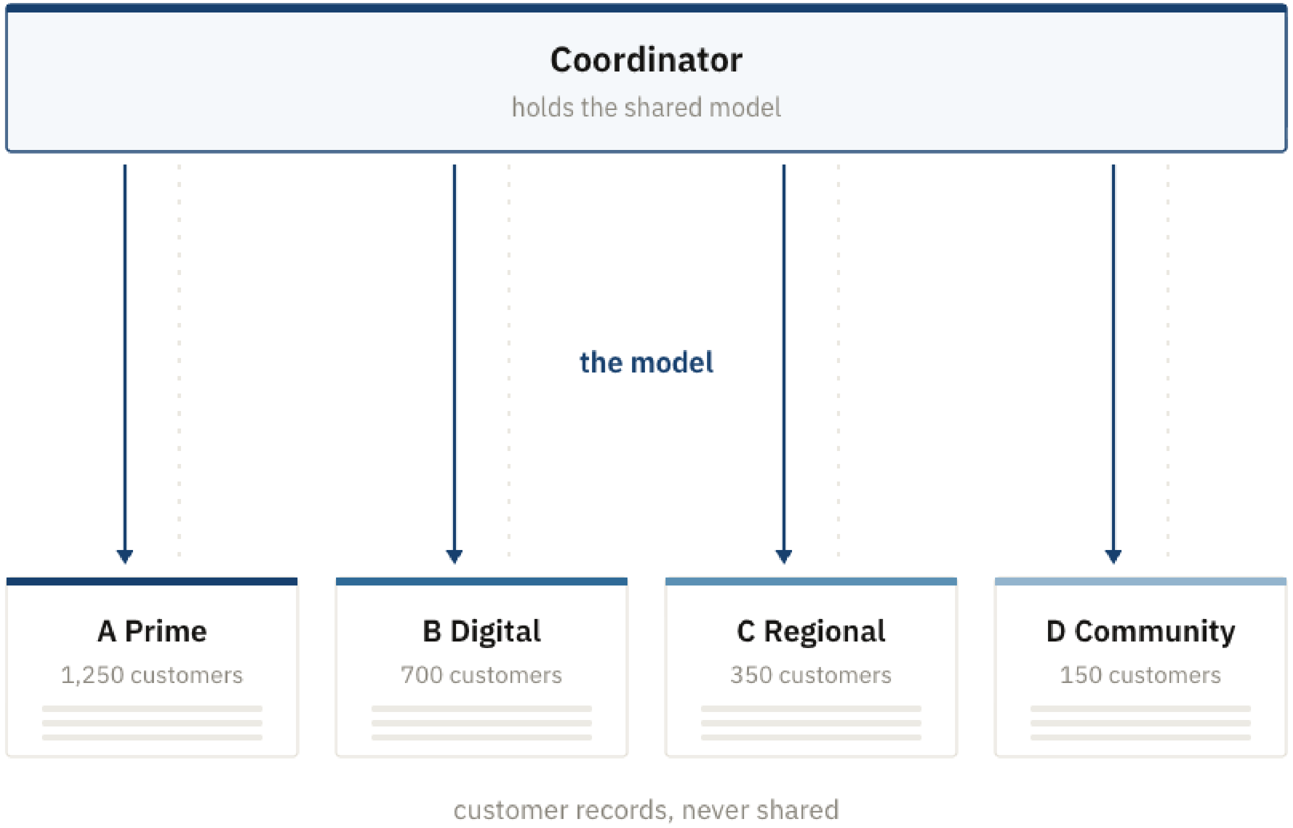


1 **Set up the federation.**

Real life examples

coordinator	clients
A research centre	**Hospitals**
A keyboard maker	**Phones**
A manufacturer	**Its factories**

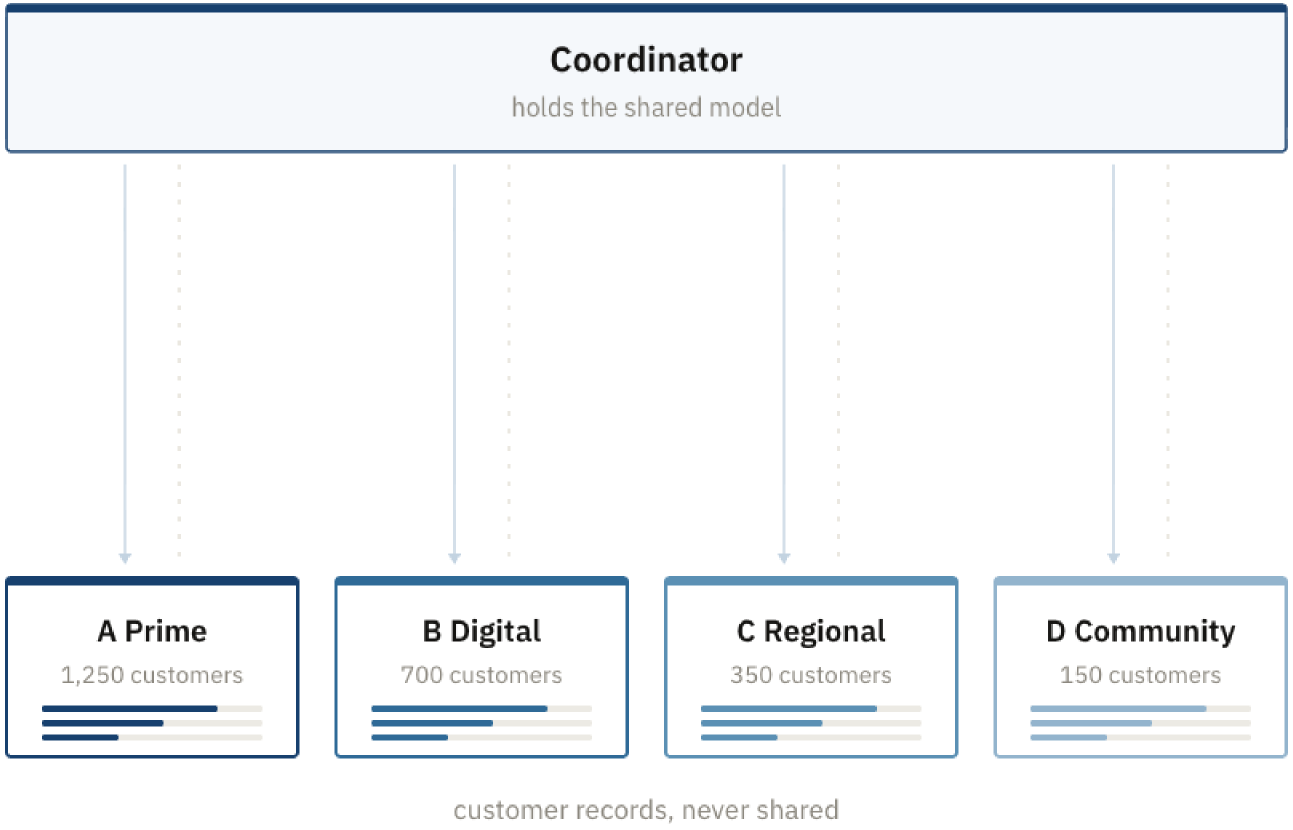
Step 2: The model goes out.



1 Set up the federation.

2 **The model goes out.**

Step 3: Each client trains at home.

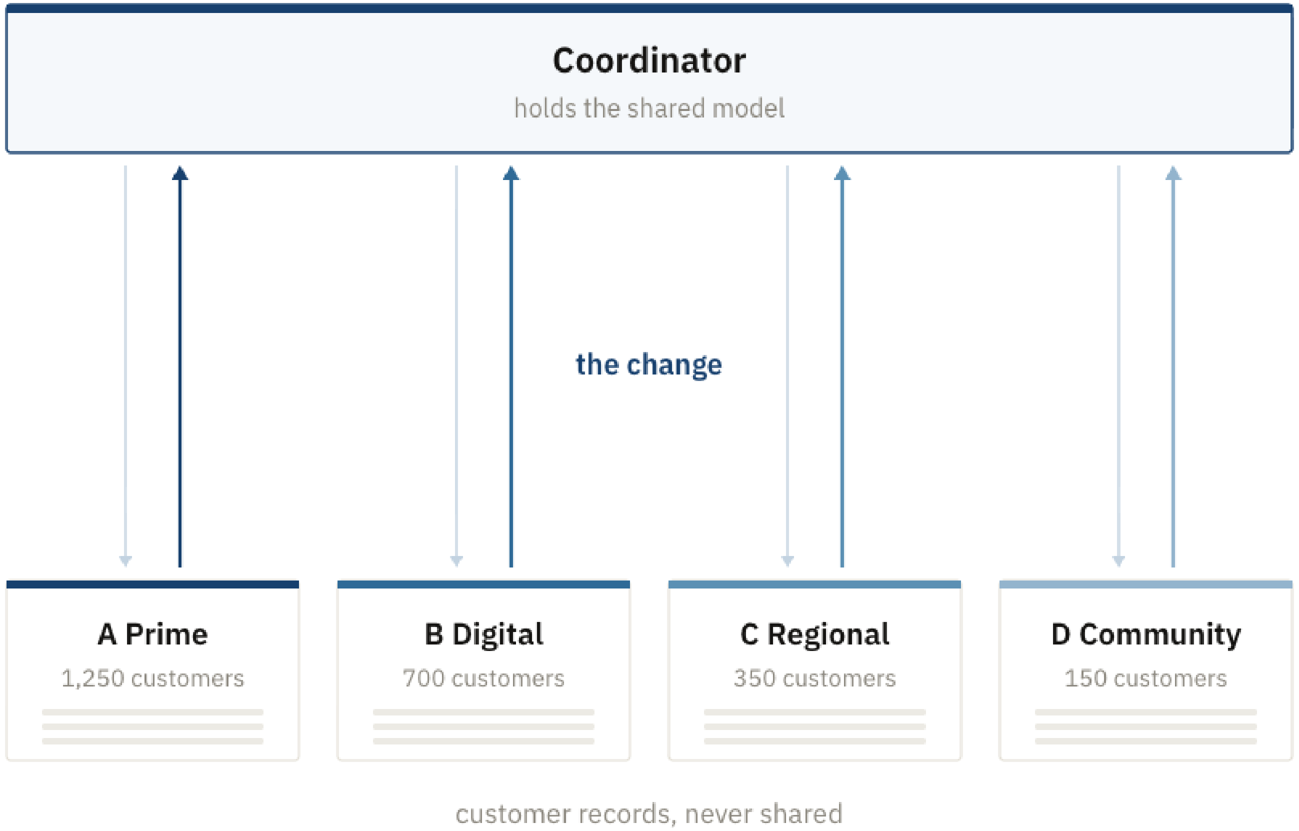


1 Set up the federation.

2 The model goes out.

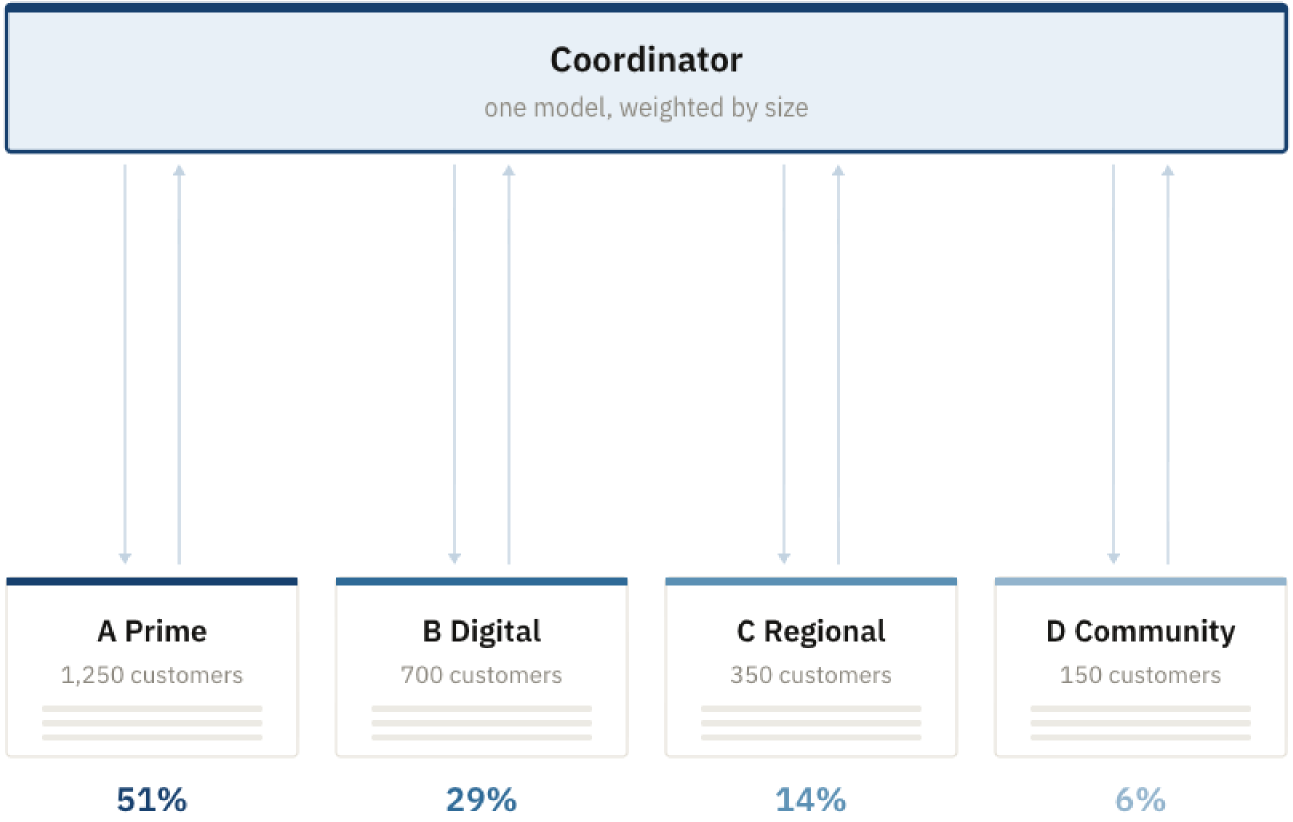
3 **Each client trains at home.**

Step 4: The change comes back.



1. Set up the federation.
2. The model goes out.
3. Each client trains at home.
4. **The change comes back.**

Step 5: The coordinator averages.



1 Set up the federation.

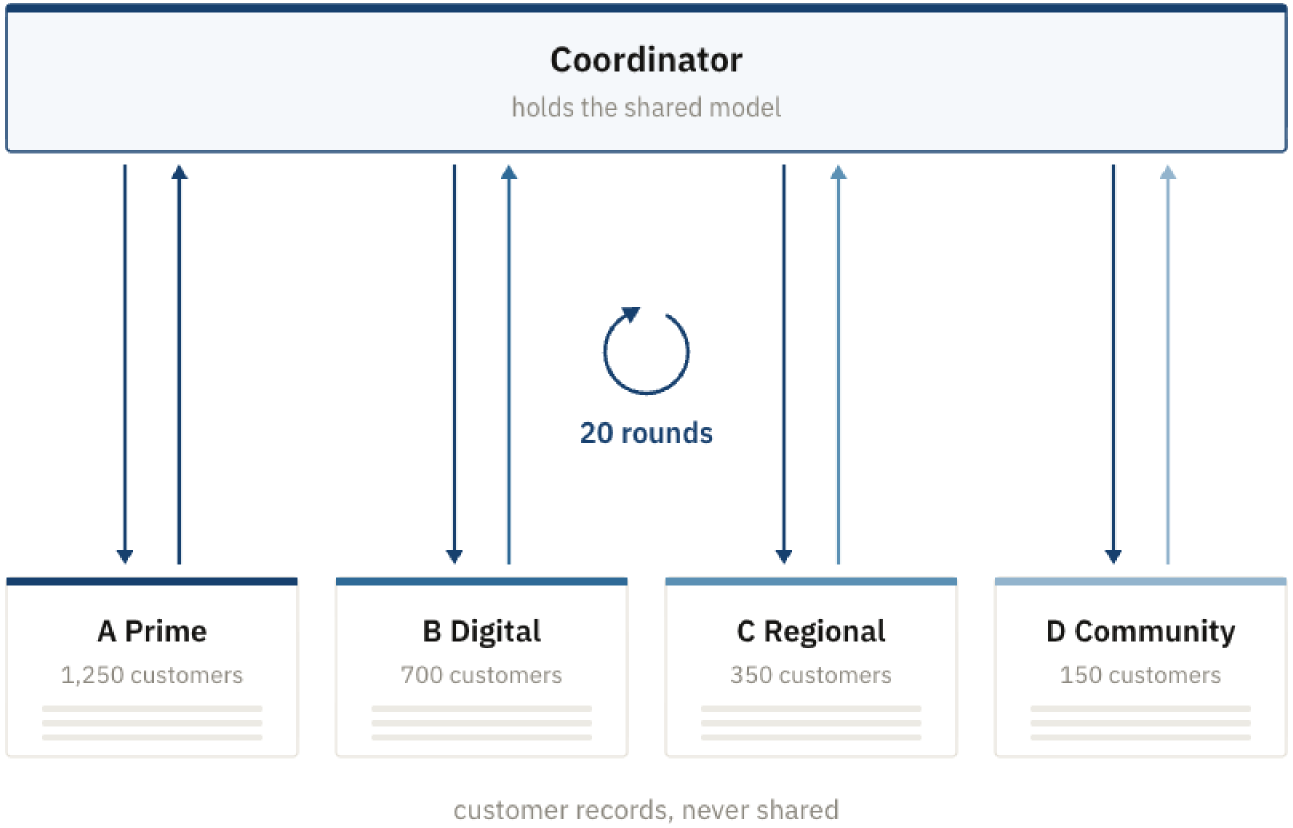
2 The model goes out.

3 Each client trains at home.

4 The change comes back.

5 The coordinator averages.

Steps 2 to 5, again and again.



1 Set up the federation.

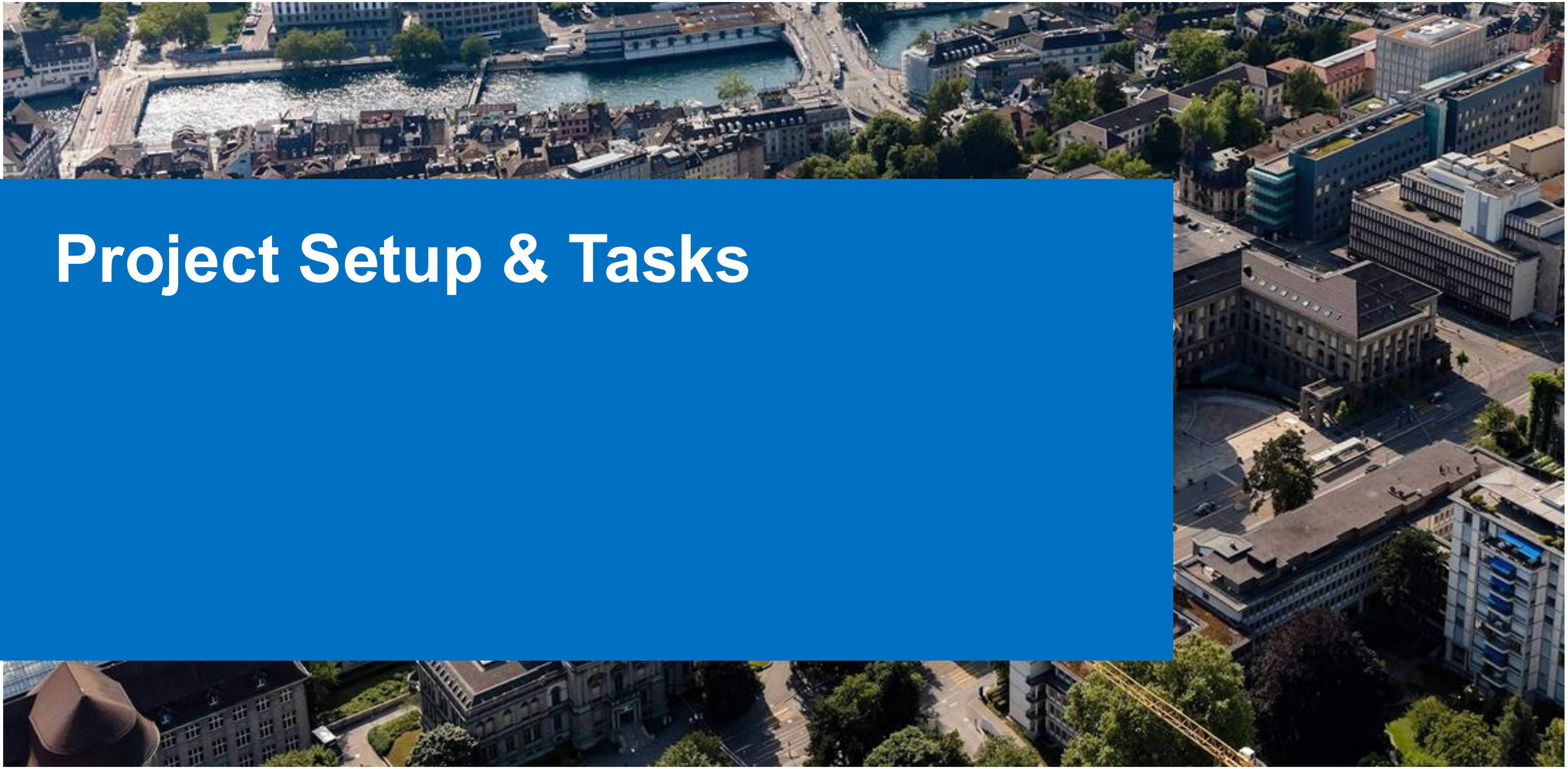
2 The model goes out.

3 Each client trains at home.

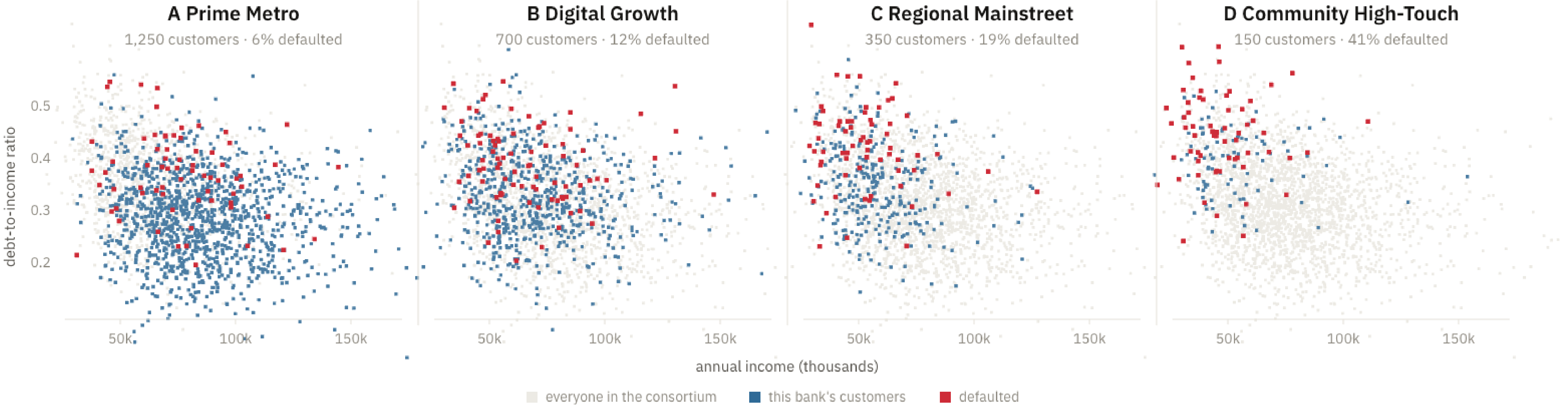
4 The change comes back.

5 The coordinator averages.

Project Setup & Tasks



Our Scenario. Our Data.



Bank	Customers	Share who defaulted	Median annual income
A Prime Metro	**1,250**	**5.5%**	**82k**
B Digital Growth	**700**	**12.3%**	**66k**
C Regional Mainstreet	**350**	**19.1%**	**54k**
D Community High-Touch	**150**	**40.7%**	**45k**
All four together	2,450	11.6%	71k

Build it, price it, attack it, defend it.

Parts 1 and 2

Motivation.

Part 3

Build it.

One model across all four. No record ever moves.

Part 4

Price it.

What it costs in rounds, local work, and bytes.

Part 5

Attack it.

Turn one update back into the customer behind it.

Part 6

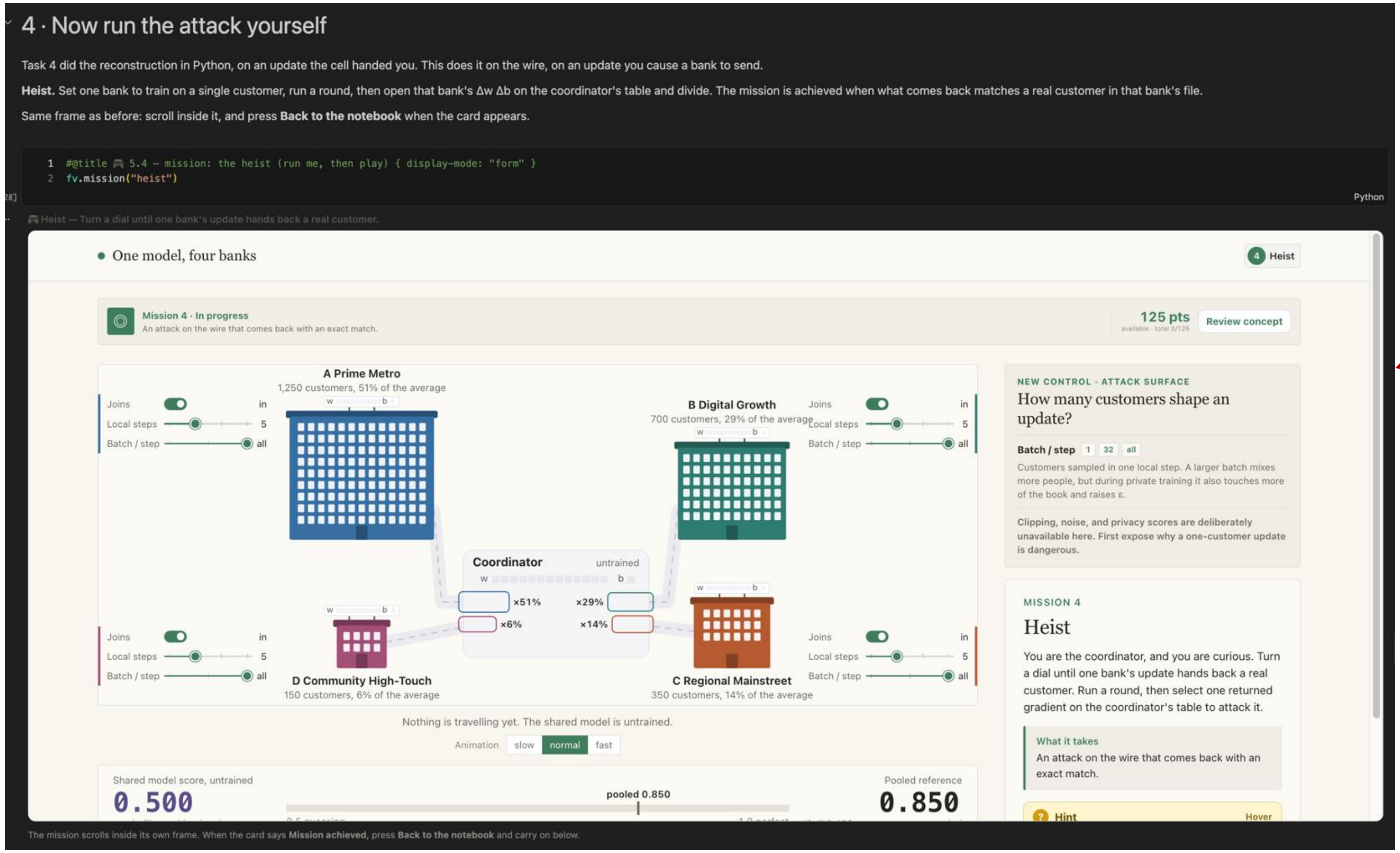
Defend it.

Clip, add noise, and price the promise that buys.

Part 7, optional

What changes when the model, the federation, or the payload gets bigger.

The hand-in is the notebook itself.



1 Fill in the code

Six tasks, twelve blanks. Run the cell and it marks each one.

2 Play the missions

Three of them, played in the game, which opens inside the notebook.

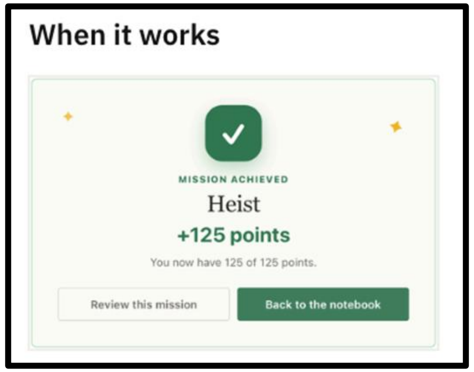
3 Fill in the tables

Five short tables, a few lines each.

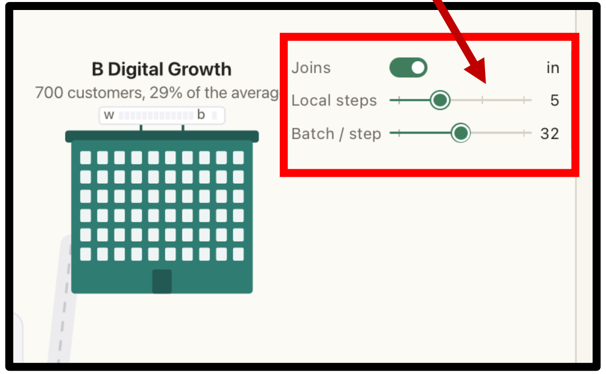
Nothing else to submit.

Mission Interface

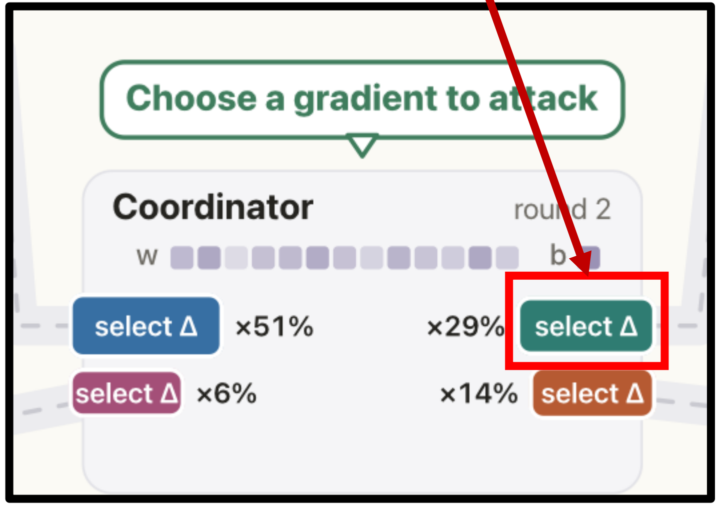
If you complete a mission, you will get this.



Play with dials.



You can select one of these.



There are hints for missions.